

Switching Concepts

Module Title: Switching Concepts

Module Objective: Explain how Layer 2 switches forward data

Topic Title	Topic Objective
Frame Forwarding	Explain how frames are forwarded in a switched network.
Switching Domains	Compare a collision domain to a broadcast domain.

Page #	Activity Type	Activity Name
2.1.4	Video	MAC Address Tables on Connected Switches
2.1.8	Activity	Switch It!
2.2.4	Check Your Understanding	Switching Domains

New Terms

Frame Forwarding

Ingress is the entry port, egress is the exit port.

The switch builds a MAC address table to forward frames on the LAN.

The switch can use either the store-and-forward or cut-through method of switch forwarding.

Switching Domains

Ethernet ports in half-duplex will be a part of a collision domain.

Full-duplex will eliminate collision domains.

A switch will flood out all interfaces except the ingress port if the frame is a broadcast or if the unicast destination MAC is unknown.

Broadcast domains may be broken up by a layer 3 device, like a router.

Switches extend broadcast domains, but can eliminate collision domains and relieve congestion.